

DESIGN EVALUATION FOR REDUCTION IN PERFORMANCE VARIATION OF TV ELECTRON GUNS

R. RANGANATHAN

TV Gun Assembly
Bharat Electronics
Jalahalli
Bangalore, India

KALYAN KUMAR CHOWDHURY and ANURAG SEKSARIA

SQC & OR Unit
Indian Statistical Institute
8th Mile, Mysore Road,
Bangalore, India

Key Words

Parametric study; Transfer functions; Taguchi methods; Taylor series expansion; Cutoff voltage.

Introduction

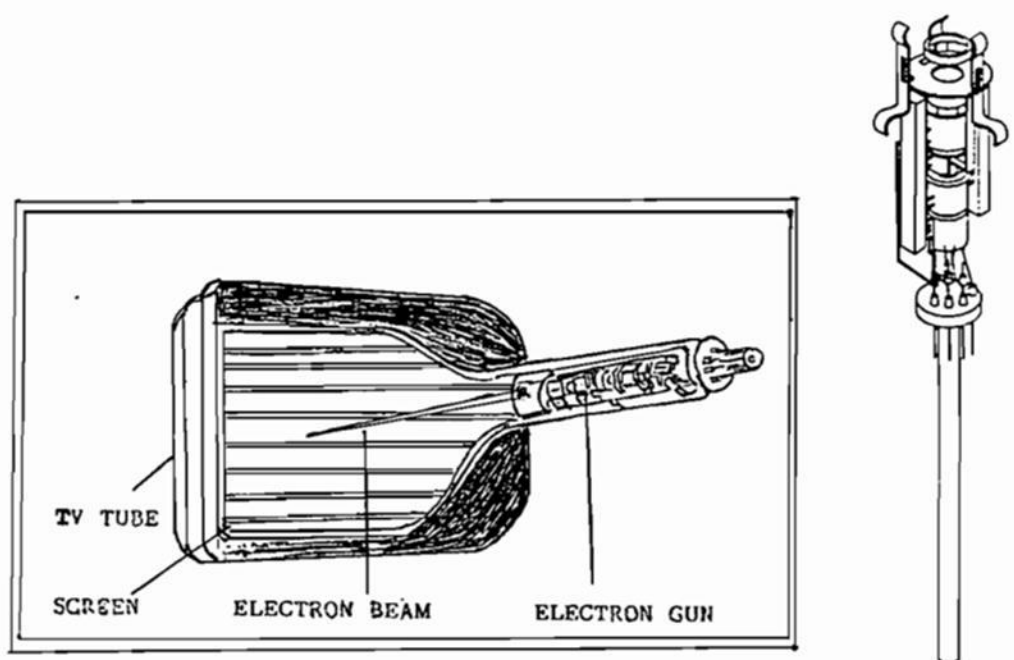
A plant manufacturing 20 inch black & white (B/W) TV picture tubes was experiencing high rejections due to high cutoff voltage (HCO) and low cutoff voltage (LCO). These were mainly attributed to deficiencies in the TV guns made in its attached captive plant using a semiautomatic process. On the other hand, a practically automatic process was used in the same gun plant for the 14 inch tubes with no complaints of this nature.

The TV electron gun is the most critical component of the picture tube which is the heart of the TV set. It receives signals through the antenna and converts them

into images seen on the TV screen. Figure 1 is a schematic diagram of an electron gun and TV tube.

All TV circuits are designed to give best picture reproduction at specified cutoff voltage of 40 to 70 volts; therefore, tubes having HCO (i.e., >70 V) will give a brighter image and tubes having LCO (i.e., <40 V) show a darker image relative to the desired level of brightness and hence cannot be used for standard TV circuits.

Such tubes are rejected and the gun is scrapped, a direct loss of Rs 55 (about US \$3) per gun. The bulb of the TV tube is reclaimed at an additional cost (about two thirds of the cost of a new tube). Before this study was undertaken, as many as 5 to 6% of guns were rejected due to HCO/LCO. The variability in cutoff voltage was analyzed statistically, and it was found that besides the process variability of ± 20 volts, there were about 2% outliers beyond the upper tolerance limit. Investigations revealed that a key component (G1 aperture) is damaged in 2% of the cases, and was hence attributed as the cause for outliers in cutoff voltage distribution. To avoid outliers, measurement and rework of G1 aperture was introduced which resulted in a reduction of rejection by 2%. Subsequent to the study, the gun plant was still incurring an annual loss of around Rs 0.6 million (about



A TV TUBE FIXED WITH ELECTRON GUN

ELECTRON GUN

Figure 1. The electron gun and TV tube.

US \$35,000) due to guns rejected for high/low cutoff voltage in addition to the additional cost incurred at the TV tube plant toward reclaiming the TV bulb.

It was reported that the cutoff voltage is influenced by several gun parameters and can be measured only after the gun is fitted inside the TV tube; thus, the measurement of cutoff voltage for experimental guns is a costly process, necessitating extensive involvement of the manufacturer of TV tubes as well as costs incurred by the incidence of rejected guns and reclaimed bulbs.

Hence, there was a need at the gun-making plant to relate the various gun parameters with cutoff voltage. In order to discover the key variables influencing cutoff and to find out their desirable tolerances without going in for any physical experimentation in the absence of this knowledge, selective assembly was resorted to, but was found ineffective.

A study was accordingly undertaken involving the use after due verification of a transfer function using the Taylor series method and Taguchi's method with the following objectives:

1. To minimize the incidence of HCO & LCO problem in 20 inch TV picture tubes
2. To identify the major parameters responsible for cutoff voltage problem
3. To avoid selective assembly

Both the methods led to the same conclusion, but Taguchi's method was found to have the following additional advantages:

It is easy to use, requiring little explanation
 No derivatives need to be generated
 Little computation is required

Analysis of Possible Causes

The schematic diagram of the TV gun is given in Figure 2. The possible cause which may affect cutoff voltage are

Test Parameters: G2 potential possibly is the only test parameter which may affect cutoff and is kept constant at the test table.

Component Dimensions: G1 aperture diameter and G1 material thickness are the two component dimensions which may affect cutoff. The nominal values and the tolerances for these parameters are given by the designer. It is known from past experience that G1 material thickness is well within the specified tolerances; hence, G1 aperture diameter was selected for the study.

Design Parameters: GK Gap & G1-G2 gap are the two design parameters which affect cutoff. The nominal values for these parameters are available but the tolerances are not known.

G1 Flatness: G1 flatness is a defect which occurs during manufacturing of TV guns and is expected to affect cutoff voltage. G1 flatness can be of two types, bulging out (i.e., +ve) or bulging in (i.e., -ve). A positive flatness will decrease

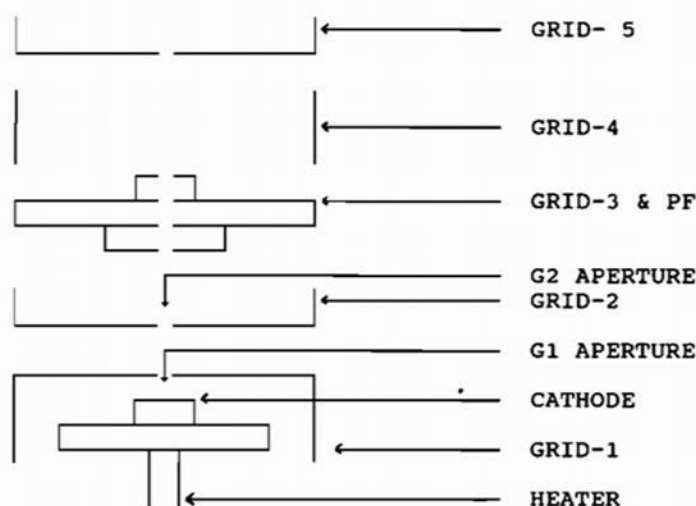
SCHEMATIC DIAGRAM

Figure 2. Schematic diagram of an electron gun.

the G1-G2 gap and increase the GK gap (as shown in Fig. 3). A good G1 component is supposed to have flatness zero. The allowable tolerance for flatness is not known and at present G1 component is segregated for different flatness values and then selectively assembled to neutralize the effect of flatness w.r.t. GK gap. For given G1 flatness, it is not feasible to adjust G1-G2 gap. The extent of the effect of G1 flatness on G1-G2 gap was not known, hence the need for study.

From the above analysis, it is clear that the effect of four gun parameters (namely G1 aperture diameter, GK gap, G1-G2 gap, and G1 flatness) on cutoff voltage need to be evaluated (others are assumed to be maintained at satisfactory level).

Investigations

A study of literature available on TV gun design revealed that the cutoff voltage can be expressed in terms of its design and test parameters by three semiempirical relationships (known as transfer function in Taguchi's terminology). Each of these relationships was tested by making a few guns with known gun parameters and the cutoff voltage measured. Only one was found to be suitable for the given design and plant condition:

$$Y = K * \frac{A^3}{(B + D)(C - D)t} * E_{c2} \quad (1)$$

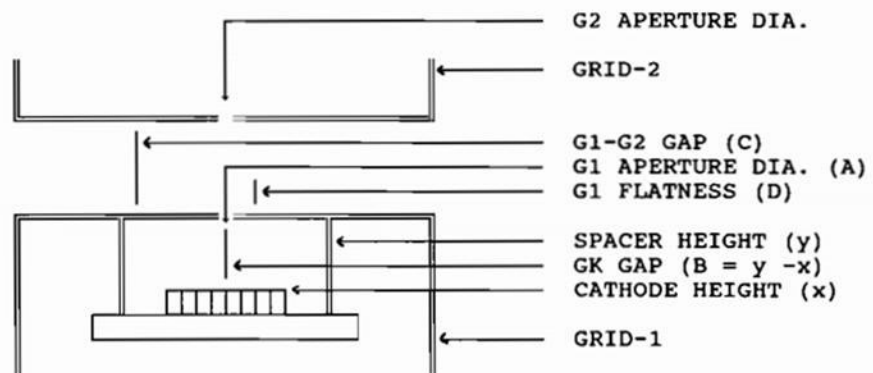


Figure 3. Effect of G1 flatness (positive) on GK gap and G1-G2 gap.

where

Y = Cutoff voltage (in volts)

K = Constant of proportionality (7.3282×10^{-3})

A = G1 Aperture diameter

B = GK Gap

C = G1-G2 Gap

D = G1 Flatness

t = G1 Material thickness, E_{c2} : G2 potential.

As discussed earlier, the effect of gun parameters A , B , C , and D are to be studied and others are kept constant; hence, the transfer function can be rewritten as:

$$Y = f(A, B, C, D) = K' \cdot \frac{A^3}{(B + D)(C - D)} \quad (2)$$

where $K' = (K \cdot E_{c2})/t$

Approach to Identify Key Gun Parameters

Here the approach was to break the total variability of cutoff voltage into a number of components corresponding to different parameters and evaluate the contribution ratio by using Taguchi's method as well as Taylor series expansion method.

Taguchi's Method

To apply Taguchi's method for design evaluation, a three-level factorial "Experimental Design" is created. The nominal value (x_{i0}) of the gun parameter is the center level and $x_{i0} \pm \sqrt{3/2}\sigma_i$ ($i = 1, 2, 3, 4$) are the high and the low level,

respectively, with σ_i as the standard deviation of the i -th gun parameter. The function $Y = f(A, B, C, D)$ is evaluated using a computer program at all $N = 3^4$ combinations of levels giving the values $Y_1, Y_2, Y_3, \dots, Y_N$. As the transfer function f is known in analytical form, the functional evaluation (Y_i 's) for each treatment combination is nonrandom in nature and contains no sampling error. Analysis of variance was then carried out on Y_i 's to evaluate contribution ratio. In Taguchi's method, the normal distribution has been replaced by a three-point discrete distribution with equal mass at the points $(x_{i0} - \sqrt{3}/2\sigma_i, x_{i0}, x_{i0} + \sqrt{3}/2\sigma_i)$ for each of the gun parameters. This distribution has mean x_{i0} and standard deviation σ_i .

Taylor Series Expansion Method

The basic idea in Taylor series expansion method for design evaluation is to linearize the functional equation using a multivariate Taylor series expansion. The moments of the output variable (i.e., cutoff voltage, R) are equal to the moments of the series expansion given below.

$$\begin{aligned} R = R_0 &+ \sum_{i=1}^4 (df/dx_i)_0 (x_i - x_{i0}) + \sum_{i=1}^4 \frac{1}{2} (d^2f/dx_i^2)_0 (x_i - x_{i0})^2 \\ &+ \sum_{i=1}^4 \sum_{j>i}^4 (d^2f/dx_i dx_j)_0 (x_i - x_{i0}) (x_j - x_{j0}) \\ &+ (\text{higher order terms}) \end{aligned} \quad (3)$$

where the partial derivatives $(df/dx_i)_0$ are evaluated at the nominal values of the gun parameters such as x_{i0} and x_{j0} , and R_0 is the value of transfer function evaluated at the nominal value of each of the gun parameters and Eq. (3) can be written as

$$R = R_0 + \sum_{i=1}^4 b_i X_i + \sum_{i=1}^4 b_{ii} X_i^2 + \sum_{i=1}^4 \sum_{j>i}^4 b_{ij} X_i X_j \quad (4)$$

where

$$X_i = (x_i - x_{i0})/\sigma_i \quad (i = 1, 2, 3, 4)$$

$$b_i = (df/dx_i)_0 \sigma_i$$

$$b_{ii} = \frac{1}{2} (d^2f/dx_i^2)_0 \sigma_i^2$$

$$b_{ij} = (d^2f/dx_i dx_j)_0 \sigma_i \sigma_j \quad (i = 1, 2, 3, 4; j > i)$$

For the given set of values of parameter, cutoff voltage was evaluated using transfer function [i.e., Eq. (2)] and the second-order Taylor series expansion [i.e., Eq. (4)]. The results from both were found to be reasonably close; hence, the expansion was truncated at second order for further use.

Using Eq. (4), the approximate mean and variance of cutoff voltage were obtained in terms of first four moments of the gun parameters and are as follows

$$\text{Mean} = R_0 + \sum_{i=1}^4 (b_{ii}) \quad (5)$$

$$\begin{aligned} \text{Variance} = & \sum_{i=1}^4 [b_i^2 \mu_{i2} + 2b_i b_{ii} \mu_{i3} + b_i^2 \mu_{i4}] \\ & + \sum_{i=1}^4 \sum_{j>i}^3 [(2b_{ij} b_{jj} + b_{ij}^2) \mu_{i2} \mu_{j2}] \end{aligned} \quad (6)$$

where μ_{ik} ($i, k = 1, 2, 3, 4$) are the k th central moment of X_i . Since the parameters x_i follow $N(x_{i0}, \sigma_i)$ and are independent and $X_i = (x_i - x_{i0})/\sigma_i$, given

$$\mu_{i1} = 0$$

$$\mu_{i2} = 1$$

$$\sqrt{\beta_{1i}} = \frac{\mu_{i3}}{\mu_{i2}^{3/2}} = \mu_{i3} = 0$$

$$\beta_{2i} = \frac{\mu_{i4}}{\mu_{i2}^2} = \mu_{i4} = 3$$

Since the mean and variance were evaluated by using Eqs. (5) and (6), to find out the contribution to the variability in cutoff voltage made by the k th parameter, its variance was set to zero one-by-one in Eq. (6) (i.e., coefficient b_k , b_{kj} and b_{kk} were set to zero). The decrement from the total variance was then the contribution for this parameter. Iteration in this way over all the parameters yielded the contribution of each.

Analysis

To estimate the variability of the parameters A, B, C, and D some randomly selected components were measured and σ_1 , σ_2 , σ_3 , and σ_4 were estimated. These values along with the levels chosen for the study are given in Table 1. The

Table 1. Variance Estimates for the Parameters Affecting Cutoff Voltage

NO.	CODE	FACTOR DESCRIPTION	S.D. $\hat{\sigma}$
1	A	G1 aperture diameter	0.00443
2	B	GK gap	0.00832
3	C	G1-G2 gap	0.02826
4	D	G1 flatness	0.00865

transfer function (2) was then evaluated at all $N = 3^4 (=81)$ combinations of levels giving the values $Y_1, Y_2, Y_3 \dots Y_N$.

Levels of each parameter: $x_{i0} - \sqrt{3/2}\sigma_i, x_{i0}, x_{i0} + \sqrt{3/2}\sigma_i$ where x_{i0} is the nominal value for the i th parameter.

Contribution Ratio Estimation

Using Taguchi's Method

Analysis of variance was carried out for 81 functional evaluations mentioned above, and the same is given in Table 2, and the response curves are given in Figure 4.

Taylor Series Expansion Method

The values for b_i, b_{ij} , and b_{ij} ($i = 1, 2, 3, 4, j > i$) were computed and are given in Table 3.

The transfer function (2) was evaluated at the nominal values for the parameters given in Table 1 and the value of R_0 was obtained. The mean and variance for cutoff voltage were then estimated using expression (5) and (6), and are given in Table 4. The contribution ratios for different parameters were calculated using the method mentioned earlier and the same are given in Table 4 along with the values obtained using Taguchi's method. It is seen that the results obtained by using the two methods are in good agreement.

Table 2. Analysis of Variance for Cutoff Voltage

PARAMETER	CODE	D.F.	S.S.	M.S.S.	$\rho\%$
G1 Aperture diameter	A_1	1	80.4383	80.4383	2.10
	A_q	1	0.0014	0.00014	—
GK gap	B_1	1	415.6866	415.6866	10.85
	B_q	1	0.3489	0.3489	0.005
G1-G2 gap	C_1	1	3289.0700	3289.0700	85.85
	C_q	1	21.5708	21.5708	0.56
G1 flatness	D_1	1	13.7456	13.7456	0.36
	D_q	1	0.2598	0.2598	0.003
Residual pooled		72	10.0447	0.1395	
		73	10.0461	0.1376	0.26
Total		80	3831.1660	47.8896	

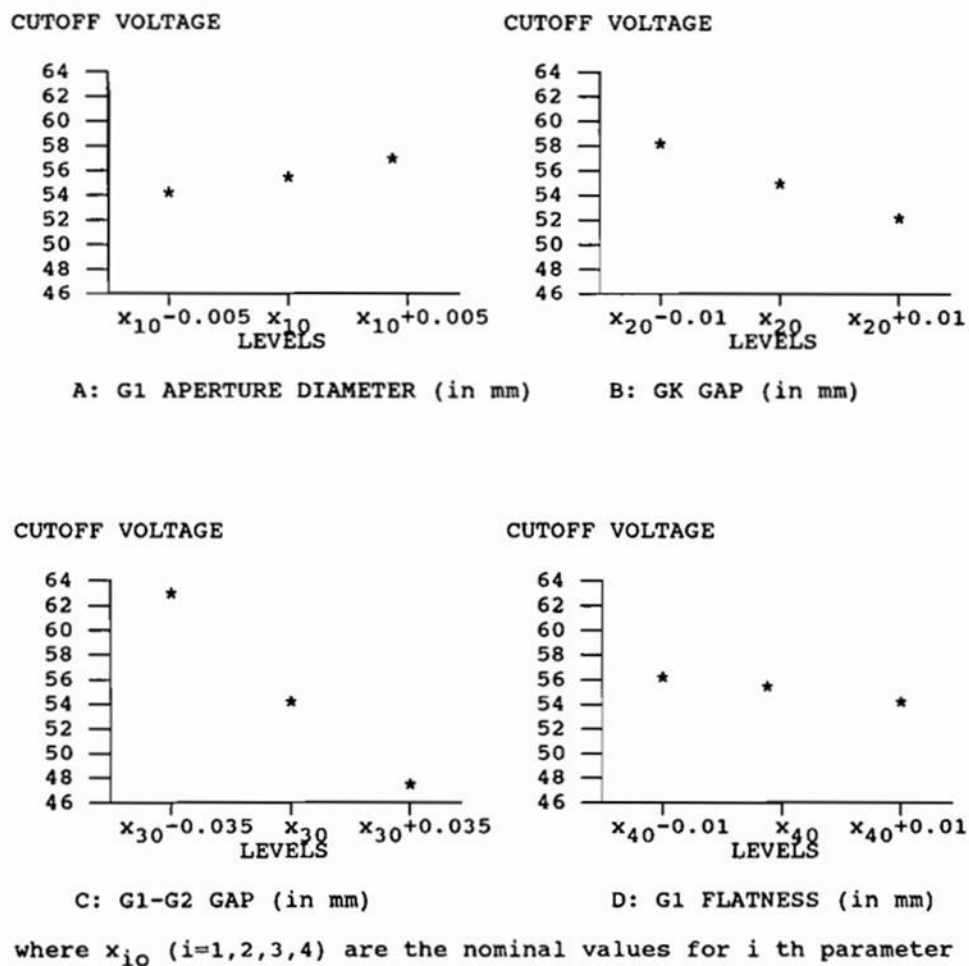


Figure 4. Response curves for different gun parameters.

Table 3. Values of b_i , b_{ii} , b_{ij}

i	b_i	b_{ii}	b_{ii}	b_{2i}	b_{3i}
1.	1.063751	.006930	—	—	—
2.	-2.264212	.094191	-.044252	—	—
3.	-6.152561	.695486	-.120246	-.191960	—
4.	-.471021	.042800	-.009206	.117567	-.159733

Table 4. Comparison of Results Obtained Using Taguchi's Technique with the Results of Taylor Series Expansion Method

PARAMETER	RESULTS	
	TAGUCHI'S METHOD	TAYLOR'S METHOD
Mean $\pm$ SD	54.42 $\pm$ 6.92	56.03 $\pm$ 6.96
Contribution by parameters		
A: G1 aperture diameter	2.10	2.50
B: GK gap	10.86	11.50
C: G1-G2 gap	86.41	86.08
D: G1 flatness	0.36	0.24

Tolerance Analysis

From Table 4, standard deviation for cutoff voltage ($\sqrt{V_T}$) = 6.92 and thus expected variation of cutoff voltage = $\pm 3\sqrt{V_T} = 20.76 \approx 21$ volts which is higher than the desired tolerance of ± 15 Volts. Further, it is to be noted that the factor C (i.e., parameter G1-G2 Gap) contributes more than 80% of the total variability.

In order to get closer tolerance the parameter C must be controlled with tighter tolerance. If the standard deviation of C is reduced by half, the new variance V_Y for cutoff voltage will become

$$\begin{aligned}
 V_Y &= V_T * [1 - \rho_C * \{1 - (1/2)^2\}] \\
 &= 47.8896 * [1 - 0.8585 * \{1 - (1/2)^2\}] \\
 &= 17.05 \\
 \sqrt{V_Y} &= 4.12 \\
 \pm 3\sqrt{V_Y} &= 12.39 \approx 13 \text{ Volts}
 \end{aligned}$$

With the existing variability in parameters B, C, and D around the nominal values given in Table 1, the expected tolerance for cutoff voltage is ± 21 volts around the nominal value of 55 volts. By squeezing the variability of parameter C by half, the expected tolerance for cutoff voltage is ± 13 volts.

Conclusions

From the above analysis it was concluded that

1. The parameter C (G1-G2 Gap) is most important for controlling variability in cutoff voltage. It contributes more than 80% of the total variability in cutoff

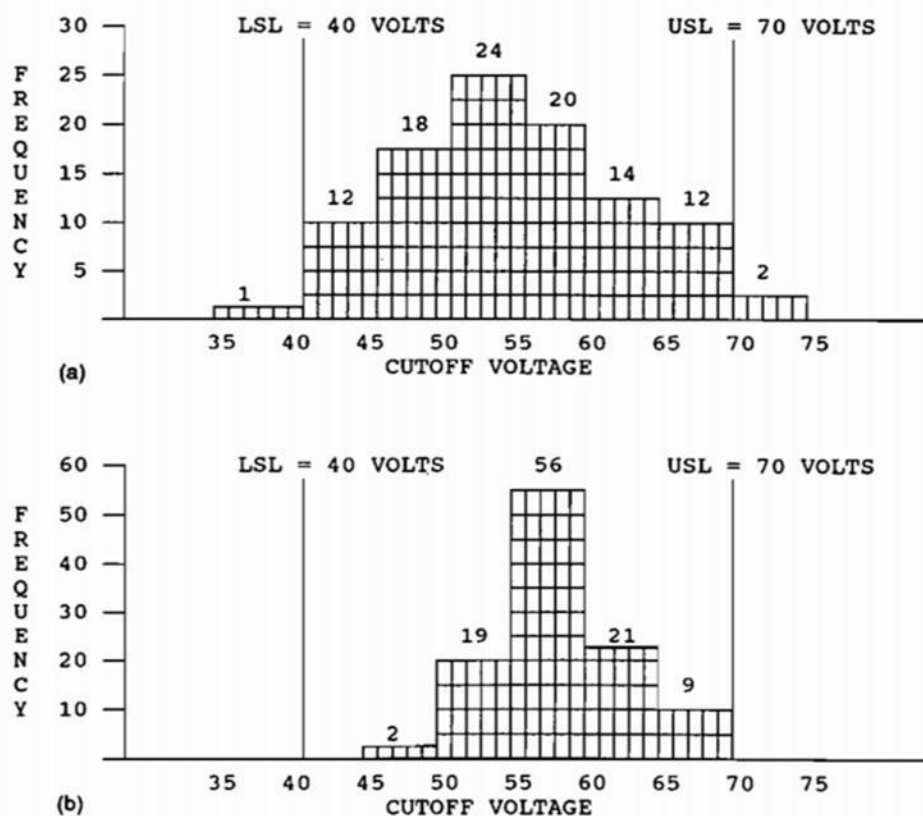


Figure 5. Histograms for cutoff voltage before and after study.

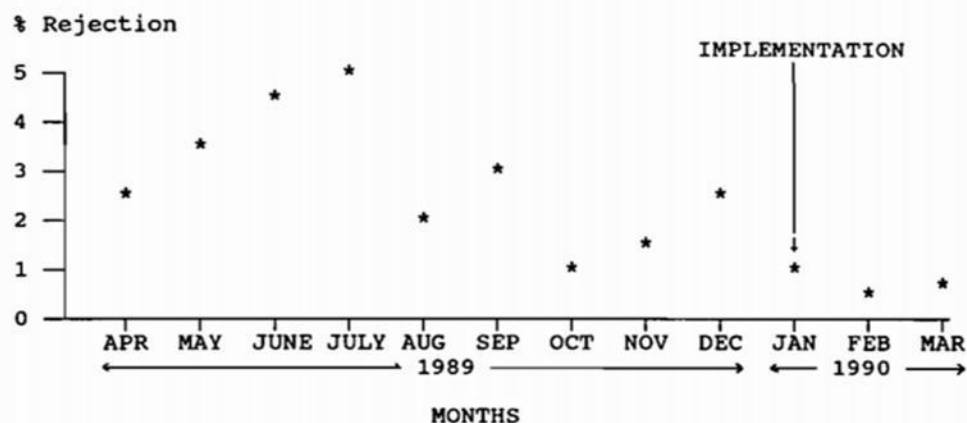


Figure 6. Monthly rejections due to HCO/LCO.

voltage (Table 4). In order to achieve the desired tolerance of ± 15 volts, the process variation in G1-G2 gap has to be reduced to the half of its existing value.

2. Parameter D (G1 flatness) is not a significant parameter for control of variability in cutoff voltage. It contributes less than 1% of the total variability in cutoff voltage (Table 4); hence, segregation and selective assembly will not make any significant improvement.

Actions Taken

To verify the results, a pilot trial was carried out which confirmed the results. The following actions were implemented on permanent basis:

1. Segregation and selective assembly for G1 flatness was stopped.
2. For controlling G1-G2 gap, process control was strengthened. This will be followed by developing a machine for consistency of G1-G2 gap.

Benefits

The benefits of the study are as follows:

1. A direct saving of Rs 0.42 million (about US \$25,000) per annum due to reduction in the spread in cutoff voltage by 25% (see Fig. 5) and hence bringing down the gun failures by 2% (see Fig. 6).
2. Elimination of additional operation of selective assembly, resulting in greater productivity.
3. Better customer satisfaction due to improved performance.
4. Facilitating the development of indigenous electron gun for "computer monitors."

Acknowledgment

The authors acknowledge with thanks the active support at every stage extended by Mr. K. M. Karumbaiah, General Manager, ET Division, Bharat Electronics, Bangalore.

References

1. Zhigarev, A., *Electron Optics & Electron Beam Devices*, MIR Publisher, Moscow.
2. Thierfelder, C. W., *Black & White Picture Tubes (682-707)*, Electron Tube Design, Radio Corporation of America, Edison, NJ, 1962.
3. Taguchi, G., *System of Experimental Design, Engineering Methods to Optimize Quality & Minimize Costs* (Vol. I & II), Unipub, Kraus International Publications, New York and American Supplier Institute, Inc., MI, 1988.

4. Phadke, M. S., *Quality Engineering Using Robust Design*, Prentice Hall, Englewood Cliffs, NJ, 1989.
5. D'errico, J. R. and Zaino Jr., N. A., *Statistical Tolerancing Using a Modification of Taguchi's Method*, *Technometrics*, 30, 397-405, 1988.
6. Cox, N. D., *The ASQC Basic References in Quality Control, Statistical Techniques*, Vol. 11, *How to Perform Statistical Tolerance Analysis*, ASQC, Milwaukee, WI, 1986.

About the Authors: Mr. R. Ranganathan passed B.E. (Hons.) from Madras University in 1966 with over 15 years of experience in semiconductors and 6 years of experience in the television tubes & electron gun manufacturing department of M/s Bharat Electronics, Bangalore. He is currently Deputy General Manager of Electron Gun Manufacturing Unit.

Mr. Kalyan Kumar Chowdhury has a Master's of Statistics (Indian Statistical Institute (ISI), 1976) and postgraduate diploma in SQC and OR (ISI, 1977). He is a senior specialist in the SQC and OR Unit of ISI, Bangalore engaged in extensive consultancy services for promoting application of quantitative methods (e.g., design of experiments, Taguchi methods, reliability engineering) and quality management concepts in a variety of industries. He is also a faculty member of various postgraduate courses in SQC and OR run by the institute, including one run for International Statistical Education Centre (ISEC). He has published in journals/conference proceedings.

Mr. Anurag Seksaria holds a Master of Statistics (Agra University, 1986) and postgraduate diploma in SQC and OR (Indian Statistical Institute (ISI), 1987). Currently he is working as a specialist in the SQC and OR Division of ISI, Pune and is engaged in consultancy services in the field of quality control and reliability.